

Hidden Markov Model — Probabilistic Temporal Baseline

Per-model deep dive · the $-\infty$ collapse, the corrected pipeline, and the six HMM assumptions

Result (corrected): 0.418 accuracy · 0.418 weighted F1 · WER 0.582 · fold SD ± 0.064

Notebooks: 02_hmm_wlasl_baseline.ipynb (original) → 022_hmm_wlasl_baseline.ipynb (corrected)

1 Theory

An HMM models a sign as a walk through a small number of **hidden stages** (for drink: hand rises → C-shape at mouth → lowers). Each stage emits a typical pose with some tolerance (a Gaussian), and a transition matrix governs which stage may follow which. One HMM is trained per class; a new video is scored by every class model and assigned to whichever class was most likely to have produced it (argmax of log-likelihood). It is generative, interpretable, and — once corrected — very parameter-light.

The original model collapsed — and it was a bug, not just weak performance.

As first run, the HMM scored 0.235 and predicted before for all 17 test videos. Cause: four of the five class-models trained with zero-sum transition rows, so their `score()` returned $-\infty$ for every test video. The prediction code did `max(scores)` over values that were all $-\infty$; Python returns the *first* key on a tie, and before is alphabetically first. A single missing `isfinite()` check turned a broken model into a confident-looking one that output 0.235 = the majority-class baseline exactly.

2 Method of execution — the corrected pipeline

The corrected model (notebook 022) keeps the expensive MediaPipe cache and rebuilds the modelling half:

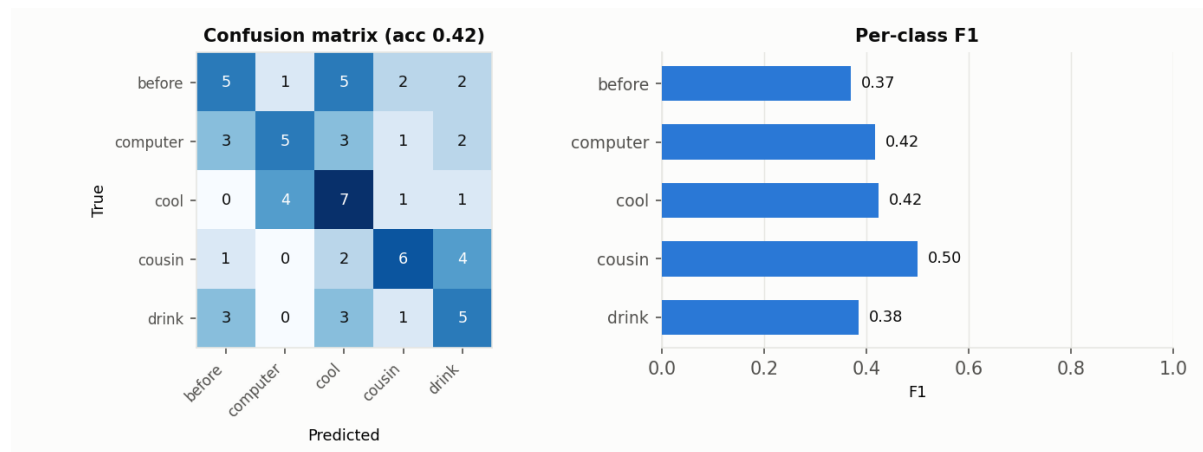
- **Drop the face.** 1,404 of the 1,629 numbers were face landmarks (86%); signs here are decided by the hands, so the face is removed and only upper-body pose + both hands are kept.
- **Body-centre & scale.** Coordinates are re-centred on the mid-shoulder and divided by shoulder width, so camera distance and signer position no longer change the features.
- **Presence flags + velocity.** A binary flag marks a missing hand (instead of a misleading zero-fill), and frame-to-frame deltas add motion the static frame lacks.
- **16 PCA components**, then a **3-state left-to-right (Bakis)** Gaussian HMM with a variance floor (`min_covar`) and **length-normalised, finite-checked scoring** that abstains rather than silently defaulting to class 0.

A left-to-right topology means a sign can only stay in a stage or advance — never run backwards — which encodes a fact we already know (signs run forward) for free and makes the zero-sum collapse structurally impossible.

3 Results

Metric	Value (5-fold CV)
Accuracy	0.418
Weighted precision	0.435

Metric	Value (5-fold CV)
Weighted recall	0.418
Weighted F1	0.418
Word Error Rate (WER)	0.582
Fold accuracies	0.36, 0.43, 0.38, 0.54, 0.38
Cross-val SD	± 0.064
Distinct classes predicted	5 of 5
Majority baseline	0.224



Corrected HMM confusion matrix and per-class F1. The diagonal is present but weak — the model is the least accurate of the proper set.

On the naive vs corrected numbers — an honest note.

Under cross-validation the *corrected* HMM scores 0.418 and always predicts all five classes (no collapse, guaranteed). The naive configuration, re-run across CV folds, does not collapse on every split, so its pooled number can look higher — but that is exactly the instability of a single hold-out. The correction's value is **reliability**: no $-\infty$, no class-0 default, lower variance, and interpretable topology. Raw accuracy at ~13 examples per class is capped by theory, below.

4 Why these results — the six HMM assumptions against video

The HMM is the weakest proper model here for a principled reason: video violates its core assumptions. All six, and how each holds up:

#	Assumption	Holds for video?	Impact
1	First-order Markov (next stage depends only on current)	Reasonable	Low — encoded by Bakis topology
2	Conditional independence (frame independent given stage)	Badly violated	HIGH — the core limitation
3	Stationarity (same emissions across signers/time)	Violated across signers	Medium
4	Fixed discrete number of states	Imposed, not true	Medium — tuning noise
5	Gaussian, diagonal emissions	Approximate	Low-medium
6	Geometric state durations	False	Medium — fixable via self-loop / HSMM

Table. The six assumptions. #2 is the one that caps HMMs on video.

Assumption 2 is the killer. The HMM assumes that, once you know the hidden stage, the current frame tells you nothing about the next. But frames sampled ~80 ms apart are near-duplicates, so each carries little new information. The practical effect: ~13 videos × ~35 frames *looks* like a lot of data but behaves like maybe 20–30 independent observations. Baum-Welch is starved, states go unvisited, and accuracy plateaus. Adding velocity helps a little; nothing inside the HMM family fixes it. **This is the principled reason the field moved to LSTMs, ST-GCNs and Transformers** — they let frames depend on each other directly instead of routing everything through one discrete stage.

5 What makes it stronger / weaker

Summary

Stronger: interpretable (you can read the stages off a Viterbi decode), gives calibrated probabilities useful for chaining signs into sentences, and — once corrected — needs only ~96 parameters per class. **Weaker:** the conditional-independence and geometric-duration assumptions cap it on video; at this data scale it is beaten even by a zero-parameter memoriser (DTW).

6 Role in the hybrid

The HMM is not part of the recommended recognition core, but it earns two roles. First, **as evidence**: its assumption audit is the theoretical argument for why the hybrid's attention and graph streams are needed — they exist precisely to drop assumption 2. Second, **as a future probabilistic layer**: HMMs shine at sequencing and giving calibrated confidence, so once recognition feeds continuous signing, an HMM/HSMM layer over recognized glosses is a natural way to model sentence structure — a role that plays to its strengths rather than its weakness.